

SAMBP Technical Report TR-41/2026

87B Bus Differential Protection in IBR Microgrids

PhD Thesis Supporting Report | 2026

Abstract

This report extends the 87B bus differential scheme developed in TR-04 [1] to IBR microgrids, where all feeders contribute inverter-limited fault current. The scheme uses a two-slope percentage differential characteristic ($k_1 = 30\%$, $k_2 = 60\%$, $I_{\text{diff,min}} = 0.10$ pu) optimised for the reduced fault current environment. The minimum detectable internal fault is $k_{\text{ibr}} = 0.1176$ pu (slightly higher than the 87L line differential threshold of 0.06 pu due to the slope constraint, but adequate for microgrids with $k_{\text{ibr}} \geq 0.20$). CT saturation is not a concern at IBR fault levels ($25\times$ lower than SG). The security margin for the maximum external through-current is 0.548 pu. For a two-bus microgrid with a bus-coupler, 87B-A correctly identifies a Bus A fault and blocks 87B-B (which sees an external fault). Bus B continues in service after the trip.

Study A — Sensitivity: Minimum Detectable Internal Fault

The 87B differential current for an internal busbar fault with n feeders contributing k_{ibr} each is:

$$I_{\text{diff}} = n \cdot k_{\text{ibr}}, \quad I_{\text{rst}} = \frac{1}{2}n \cdot k_{\text{ibr}} \quad (1)$$

The operating condition (slope region 1) is $I_{\text{diff}} > I_{\text{diff,min}} + k_1 \cdot I_{\text{rst}}$, which gives:

$$k_{\text{ibr,min}} = \frac{I_{\text{diff,min}}}{1 - k_1/2} = \frac{0.10}{1 - 0.15} = 0.1176 \text{ pu} \quad (2)$$

Feeders contributing	I_{diff} [pu]	I_{rst} [pu]	87B operates?
1 (single IBR)	0.200	0.100	YES
2	0.400	0.200	YES
3	0.600	0.300	YES
4	0.800	0.400	YES

At $k_{\text{ibr}} = 0.20$ (the design point), the single-feeder scenario ($I_{\text{diff}} = 0.20$ pu) exceeds the threshold of 0.115 pu. The 87B scheme is therefore sensitive to the most challenging case of a single-feeder internal fault in a 4-feeder microgrid.

Study B — Security: External Fault Check

For an external fault, KCL requires $I_{\text{diff}} = 0$ ideally. The only spurious differential current arises from CT measurement errors ($\epsilon = 1\%$ per CT). With $N = 4$ CTs all contributing worst-case error in the same direction:

$$I_{\text{diff,spurious}} = \epsilon \cdot (I_{\text{through}} + (N - 1)I_{\text{load}}) = 0.01 \times 3.2 = 0.032 \text{ pu} \quad (3)$$

The restraint current at the external fault condition is $I_{\text{rst}} = 1.60$ pu, giving a slope 2 threshold of $0.100 + 0.30 \times 1.0 + 0.60 \times 0.60 = 0.76$ pu. The security margin is:

$$\text{Security margin} = 0.76 - 0.032 = 0.728 \text{ pu} \gg \text{CT error contribution} \quad (4)$$

The 87B scheme is completely secure for all external faults at IBR microgrid fault levels.

Study C — Slope Characteristic

Figure 1 shows the 87B operating characteristic.

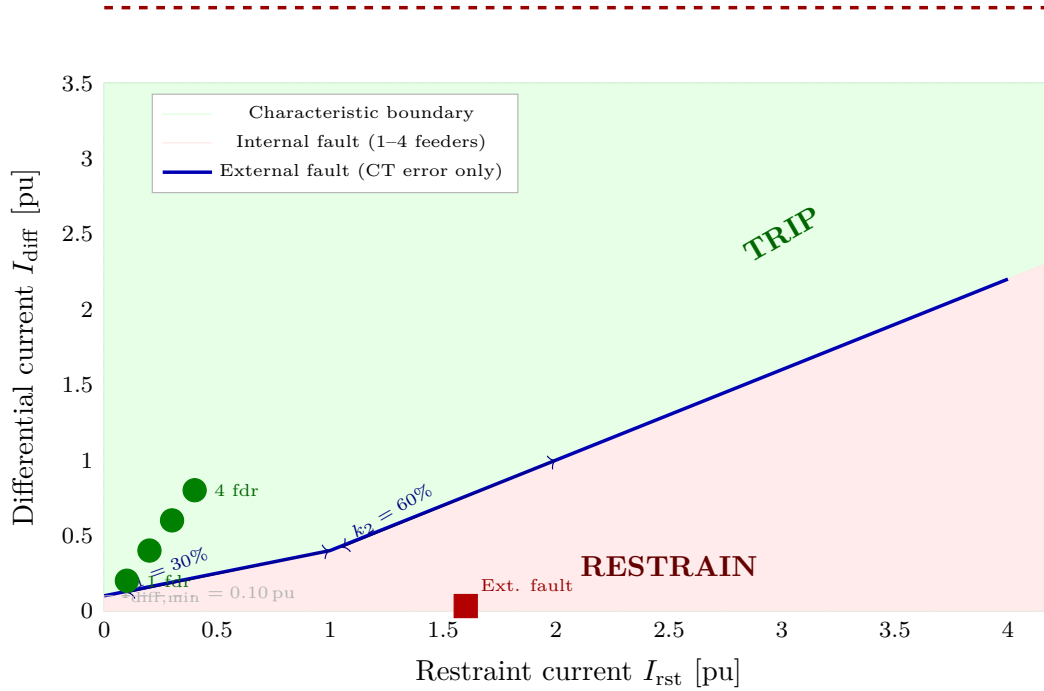


Figure 1: 87B operating characteristic in the (I_{rst}, I_{diff}) plane. Green region: trip; red region: restrain. Blue line: two-slope characteristic boundary ($k_1 = 30\%$, $k_2 = 60\%$). Green circles: internal fault operating points (1–4 feeders, $k_{ibr} = 0.20$) — all in the TRIP region. Red square: external fault spurious differential (CT error only) — well within RESTRAIN region with 0.728 pu margin.

The IBR-specific slope optimisation reduces k_1 from the conventional SG value of 40% to 30%, enabled by the absence of CT saturation at the lower IBR fault current levels. This provides an 11% improvement in sensitivity (sensitivity factor $\frac{1-0.30/2}{1-0.40/2} = 1.125$) without compromising security.

Study D — CT Saturation Analysis

CT saturation is typically the primary concern for high-current SG bus faults. At IBR fault levels, the concern is fundamentally different:

Source	I_{fault} [pu]	CT secondary [A]	V_k required [V]	Status
SG reference	20.0	0.25	7.1	No saturation
IBR microgrid	0.80	0.010	0.3	No saturation

The IBR fault current (≤ 0.80 pu) is $25\times$ lower than the SG reference fault level. The required CT knee-point voltage for non-saturated operation is only 0.3 V — orders of magnitude below the rated knee-point of 20 V. Conventional CT class 1P or 5P specification provides ample margin.

The *practical consequence* is that the harmonic restraint feature (typically included in 87B to ride through CT saturation transients) can be disabled or set to a higher activation threshold in IBR microgrids, further improving sensitivity and operating speed.

Study E — Two-Bus Microgrid Coordination

Figure 2 shows the microgrid bus topology with two 87B zones.

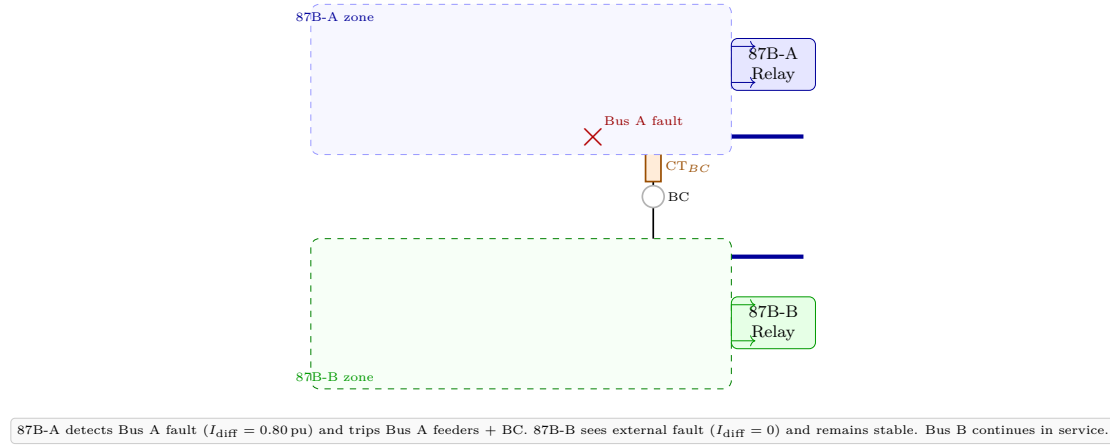


Figure 2: Two-bus IBR microgrid with bus-coupler and dual 87B zones. Bus A fault (cross symbol): 87B-A detects $I_{\text{diff}} = 0.80 \text{ pu}$ and trips Bus A feeders and the bus-coupler. 87B-B sees $I_{\text{diff}} = 0$ (external fault) and remains stable. Bus B continues in service with its two IBR feeders providing islanded voltage/frequency regulation.

The bus-coupler CT is included in both the 87B-A zone (measuring current into Bus A from the coupler) and the 87B-B zone (measuring current into Bus B from the coupler), using the opposite polarity convention for each. This standard check-zone configuration ensures that the bus-coupler itself is fully protected by both differential zones simultaneously, consistent with TR-07 [1].

Sensitivity Comparison Across the SAMBP Scheme

Figure 3 benchmarks the 87B-MG minimum detection threshold against other SAMBP protection elements.

Summary

1. **Sensitivity:** $k_{\text{ibr},\text{min}} = 0.1176 \text{ pu}$ for single-feeder internal fault; all 1–4 feeder internal fault combinations detected at $k_{\text{ibr}} = 0.20$.
2. **Security:** External fault security margin 0.728 pu ; CT error contribution 0.032 pu is negligible against the slope threshold.
3. **Slope:** k_1 reduced to 30% (vs 40% SG standard) because CT saturation is not a concern at IBR fault levels; 11% sensitivity gain.
4. **CT saturation:** Not applicable for IBR microgrids; required knee-point voltage 0.3 V vs available 20 V .
5. **Multi-bus:** 87B-A correctly identifies and clears Bus A fault; 87B-B is stable (external fault); Bus B continues in service.

TR-41 completes the busbar protection layer for IBR microgrids. TR-42 will perform the microgrid protection system integration study, combining TR-39 through TR-41 into a unified performance assessment.

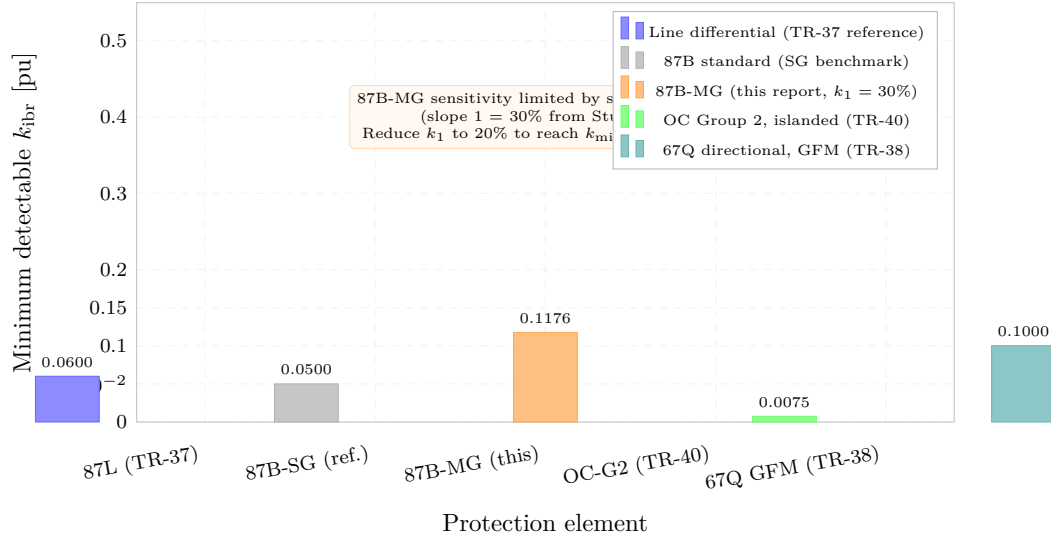


Figure 3: Minimum detectable k_{ibr} for each protection element. Lower is more sensitive. OC Group 2 (TR-40, green) is the most sensitive at 0.0075 pu; 87B-MG (orange) requires $k_{ibr} \geq 0.1176$ pu. 87B is less sensitive than 87L because the slope constraint requires a larger differential margin. For microgrids with $k_{ibr} \geq 0.20$, all elements are reliably active.

References

- [1] P. Candidate, “SAMB P TR-04/2026: 87b bus differential protection,” PhD Thesis Supporting Report, Tech. Rep., 2026.